HK IMO Training 2018-2019
Problem Set 8
Date due: 17 November 2018

Problem 33. Three circular arcs w_1, w_2, w_3 with common endpoints A and B are on the same side of the line AB; w_2 lies between w_1 and w_3. Two rays emanating from B intersect these arcs at M_1, M_2, M_3 and K_1, K_2, K_3, respectively. Prove that

$$\frac{M_1M_2}{M_2M_3} = \frac{K_1K_2}{K_2K_3}.$$

Problem 34. Let the points D and E lie on the sides BC and AC, respectively, of $\triangle ABC$, satisfying $BD = AE$. The line joining the circumcentres of $\triangle ADC$ and $\triangle BEC$ meets the lines AC and BC at K and L, respectively. Prove that $KC = LC$.

Problem 35. Let m and n be relatively prime positive integers. Determine all possible values of

$$\gcd(2^m - 2^n, 2^{m^2+mn+n^2} - 1).$$

Problem 36. A directed graph does not contain directed cycles. The number of edges in any directed path does not exceed 99. Prove that it is possible to colour the edges of the graph in 2 colours so that the number of edges in any single-coloured directed path in the graph will not exceed 9.